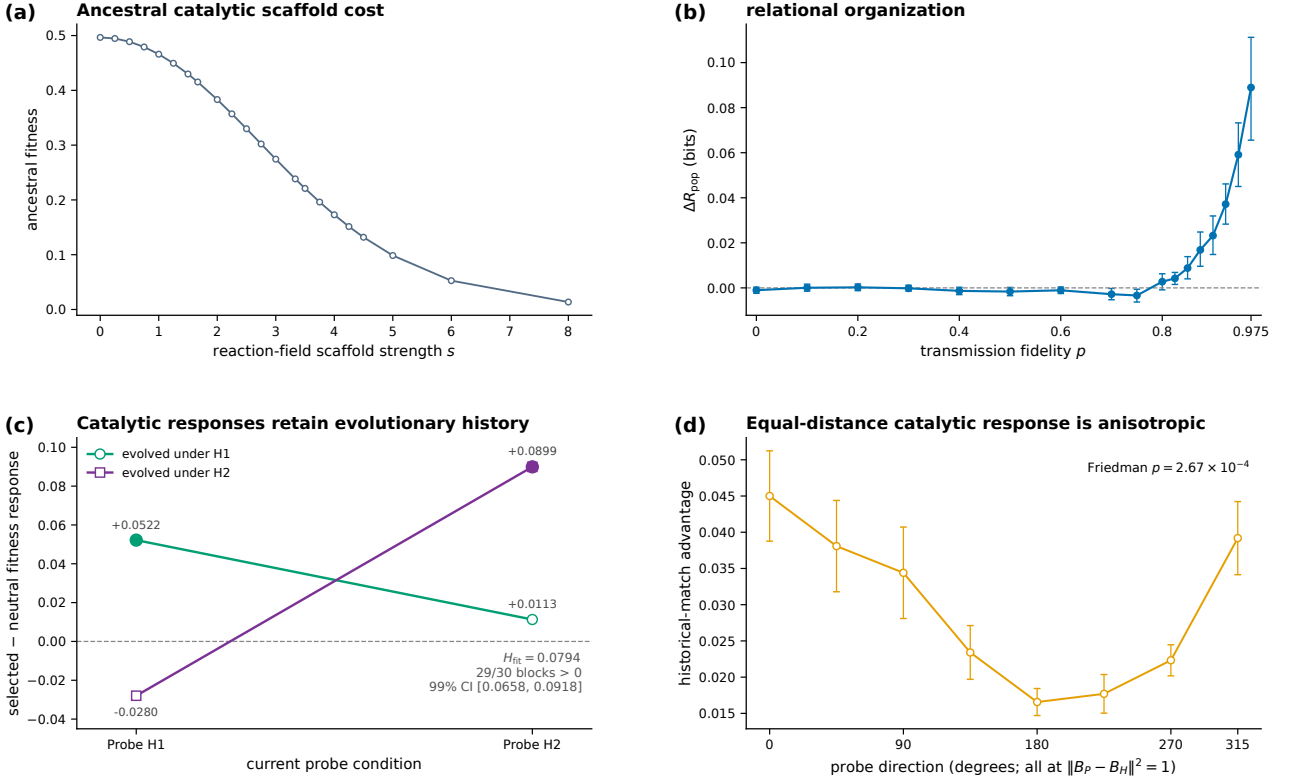

(a)
Ancestral catalytic scaffold cost
ancestral fitness
reaction-field scaffold strength s
(b)
Reliable heredity amplifies catalytic relational organization
ΔR_pop (bits)
transmission fidelity p
(c)
Catalytic responses retain evolutionary history
evolved under H1
evolved under H2
+0.0899
+0.0522
+0.0113
-0.0280
H_fit = 0.0794
29/30 blocks > 0
99% CI [0.0658, 0.0918]
selected − neutral fitness response
Probe H1
Probe H2
current probe condition
(d)
Equal-distance catalytic response is anisotropic
Friedman p = 2.67 × 10^−4
historical-match advantage
probe direction (degrees; all at ‖B_P − B_H‖² = 1)